

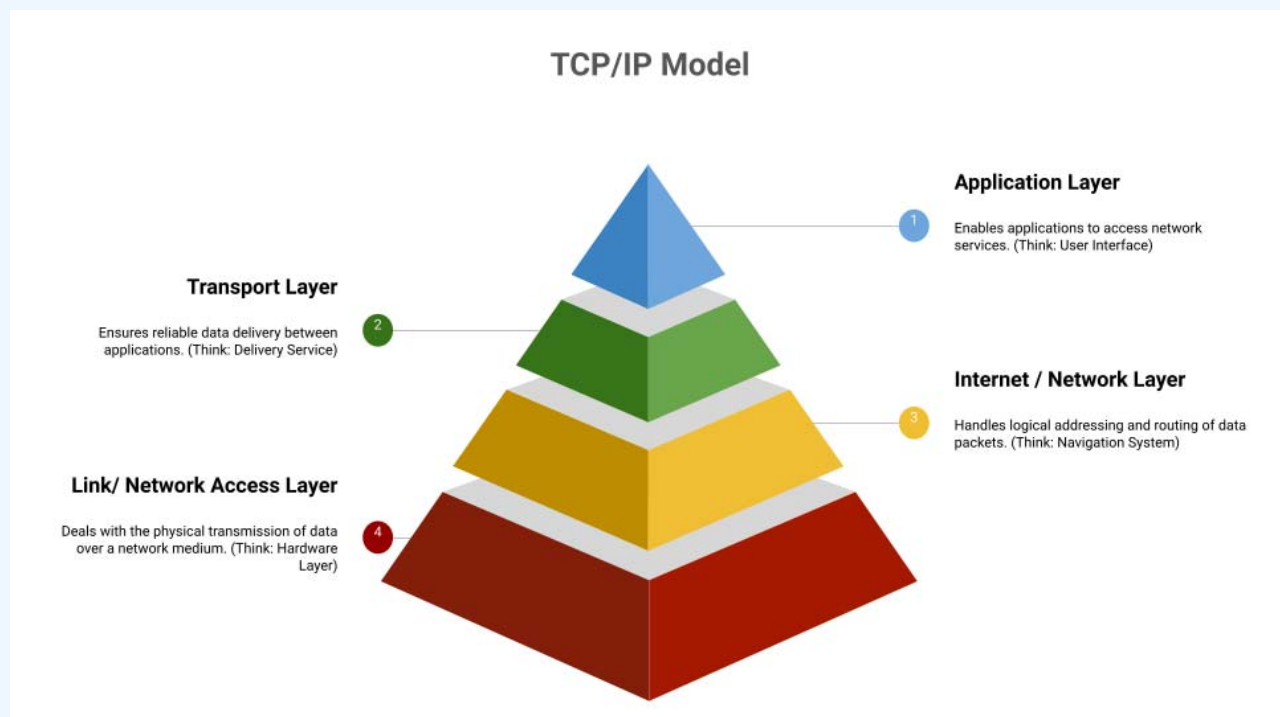
A 2.1.4 Network Protocols



A 2.1.4 Describe the network protocols used for transport and application.

Network protocols are the languages of the digital world, enabling devices to communicate. TCP ensures reliable data delivery, while UDP prioritises speed. HTTP and HTTPS govern web communication, with HTTPS adding encryption for security. DHCP automates IP address assignment, simplifying network management. Understanding these protocols is key to grasping how data flows seamlessly across networks.

What are the network protocols used for transport and application?



Just like humans use languages to communicate, computers use protocols to exchange information over a network.

These protocols are sets of rules and standards that govern how data is formatted, transmitted, and interpreted. In this section, we'll explore some essential protocols used at the Transport and Application layers of the TCP/IP model.

Network protocols are essential because they provide a common set of device rules and standards. This ensures that devices from different manufacturers and with varying capabilities can communicate and exchange information seamlessly. Without protocols, the network would be chaotic and incompatible.

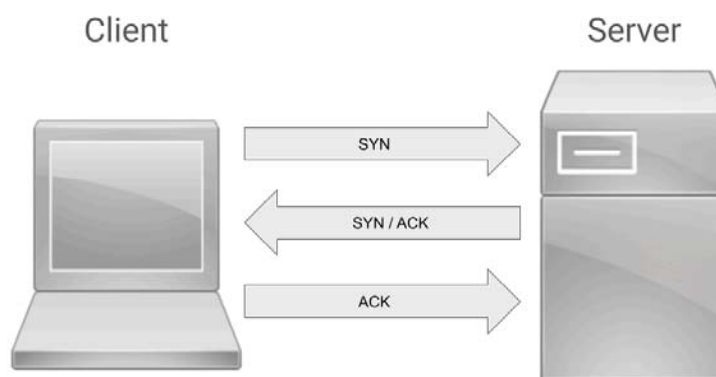
What is the purpose of protocols such as TCP, UDP, HTTP, HTTPS, and DHCP?

Transport Layer Protocols

TCP (Transmission Control Protocol)

TCP is like a reliable courier service. It ensures data is delivered accurately and in the correct order. It establishes a connection between two devices, breaks data into packets, numbers them, and checks for errors upon arrival. If packets are lost or corrupted, TCP retransmits them. This makes TCP ideal for applications where data integrity is crucial, such as web browsing, file transfer, and email.

TCP uses a three-way handshake to establish a reliable connection. The connection is full duplex (data can travel in both directions simultaneously), and both sides synchronise (SYN) and acknowledge (ACK) each other. The exchange of these four flags is performed in three steps—SYN, SYN-ACK, and ACK—as shown below



UDP (User Datagram Protocol)

UDP is like sending a letter without a tracking number. It's faster than TCP because it doesn't establish a connection or guarantee delivery. It simply sends data packets without checking for errors or reordering them. This makes UDP suitable for applications where speed is more

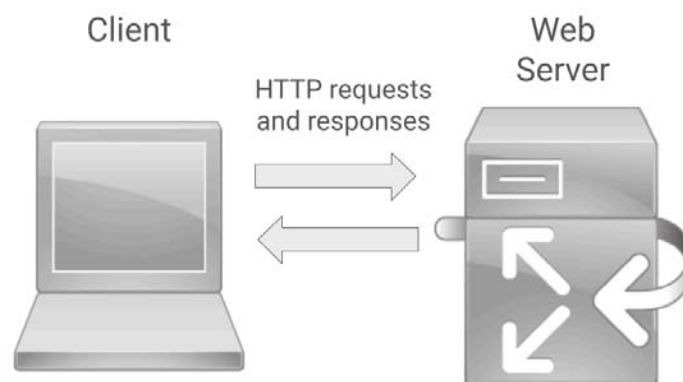
important than guaranteed delivery, such as video streaming, online gaming, and DNS lookups.

Feature	TCP	UDP
Reliability	High	Low
Order	Guaranteed	Not Guaranteed
Speed	Slower	Faster
Overhead	Higher	Lower
Applications	Web browsing, file transfer, email	Video streaming, gaming, DNS

Application Layer Protocols

HTTP (Hypertext Transfer Protocol)

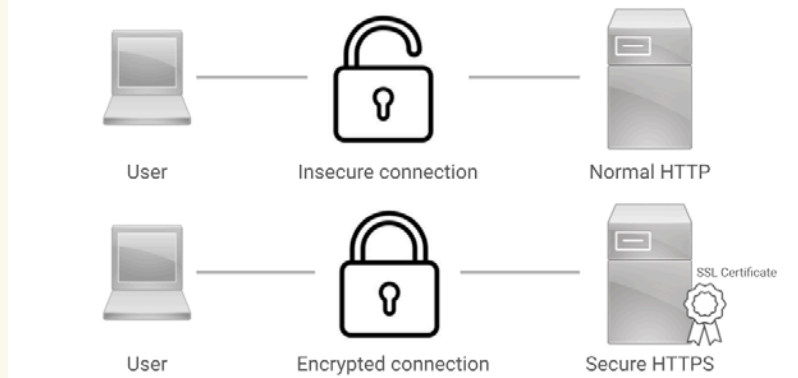
HTTP is the foundation of the World Wide Web. It's the protocol used to transfer files, such as web pages and other content, over the internet. When you type a URL in your browser, it uses HTTP to request that web page from the server.



HTTPS (Hypertext Transfer Protocol Secure)

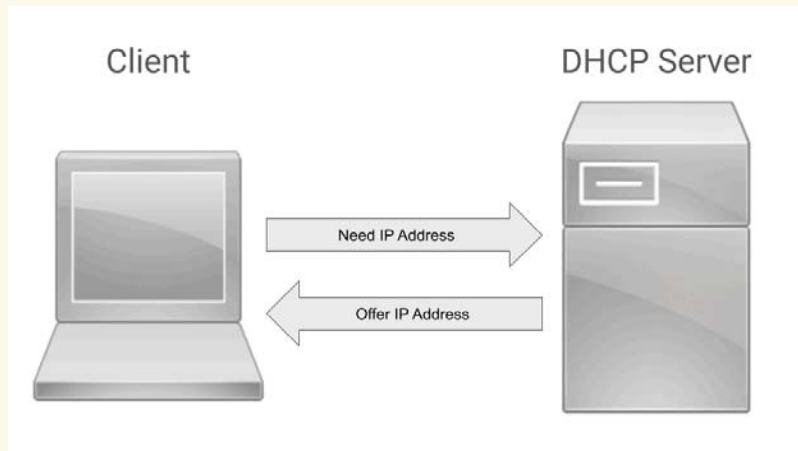
HTTPS is a secure version of HTTP. It encrypts communication between your browser and the server, protecting sensitive information like passwords and credit card details from eavesdropping and tampering. Look for the padlock icon in your browser's address bar to ensure you're using HTTPS.

HTTP vs HTTPS



DHCP (Dynamic Host Configuration Protocol)

Imagine if every device on a network needed a manually assigned IP address. That would be a nightmare! DHCP automates this process. When a device connects to a network, DHCP assigns it a unique IP address, allowing it to communicate with other devices.



Key Questions

Can you answer these key questions? Click on the eye to see more.

Explain the key differences between TCP and UDP. Provide examples of applications that use each protocol.

TCP (Transmission Control Protocol):

- Connection-oriented: Establishes a connection between devices before transmitting data.
- Reliable: Guarantees data delivery in the correct order and without errors.

- **Ordered:** Packets are numbered and reassembled in the correct sequence.
- **Slower:** The overhead of establishing a connection and ensuring reliability adds some latency.
- **Examples:** Web browsing (HTTP), file transfer (FTP), email (SMTP)

UDP (User Datagram Protocol):

- **Connectionless:** Sends data packets without establishing a connection.
- **Unreliable:** Doesn't guarantee delivery or correct order of packets.
- **Unordered:** Packets may arrive out of order.
- **Faster:** Lower overhead due to no connection establishment or error checking.
- **Examples:** Video streaming, online gaming, DNS lookups, VoIP

How does HTTP enable the transfer of web pages and other content over the internet?

HTTP defines the format of requests and responses between web browsers and servers. When you type a URL in your browser, it sends an HTTP request to the server hosting that website. The server responds with the requested web page, also formatted according to HTTP. This standardised format allows for the exchange of various content, including text, images, videos, and other multimedia.

Why is HTTPS important for secure online transactions?

HTTPS encrypts the communication between your browser and the server, preventing eavesdropping and tampering. This is crucial for protecting sensitive information like:

- **Login credentials:** Usernames and passwords
- **Financial information:** Credit card details, bank account numbers
- **Personal data:** Names, addresses, and other identifying information

How does DHCP simplify network administration and device configuration?

DHCP automates the process of assigning IP addresses to devices on a network. This eliminates the need for manual configuration and reduces the risk of IP address conflicts. When a device connects to the network, it sends a DHCP request, and the DHCP server assigns it a unique IP address, subnet mask, and other network configuration parameters.

What are the advantages and disadvantages of using TCP versus UDP?

Feature	TCP	UDP
Reliability	High	Low
Order	Guaranteed	Not Guaranteed
Speed	Slower	Faster

Feature	TCP	UDP
Overhead	Higher	Lower
Applications	Web browsing, file transfer, email	Video streaming, gaming, DNS

How does encryption ensure data confidentiality and integrity in HTTPS?

HTTPS uses SSL/TLS (Secure Sockets Layer/Transport Layer Security) to encrypt data. This encryption:

- Confidentiality: Ensures that only the intended recipient can read the data.
- Integrity: Verifies that the data has not been tampered with during transmission.

What are the potential security risks associated with using HTTP?

Eavesdropping: Data transmitted over HTTP is unencrypted, making it vulnerable to interception.

Man-in-the-middle attacks: Attackers can intercept and modify data in transit.

Data tampering: Data can be altered without detection.

How does DHCP help manage IP addresses in a network?

DHCP centralises IP address management, preventing conflicts and simplifying administration. It allows for:

- Dynamic allocation: IP addresses are assigned automatically as devices connect.
- Efficient use of IP addresses: Addresses can be reused when devices disconnect.
- Centralised configuration: Network settings can be managed from a central server.

What are some other application layer protocols besides HTTP and HTTPS?

- FTP (File Transfer Protocol): For transferring files between devices.
- SMTP (Simple Mail Transfer Protocol): For sending email.
- POP3 (Post Office Protocol version 3) and IMAP (Internet Message Access Protocol): For receiving email.
- DNS (Domain Name System): For translating domain names (e.g., [invalid URL removed]) to IP addresses.
- SSH (Secure Shell): For secure remote access to devices.

Quiz

Which protocol is responsible for reliable data transmission, ensuring packets arrive in the correct order?

- A. ☐ UDP
- B. ☐ DHCP
- C. ☒ TCP
- D. ☐ HTTP

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Which protocol is best suited for applications where speed is more important than guaranteed delivery, like video streaming?

- A. ☐ TCP
- B. ☐ HTTPS
- C. ☒ UDP
- D. ☐ DHCP

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What is the primary function of HTTP?

- A. ☐ Ensuring reliable data delivery
- B. ☐ Encrypting data transmission
- C. ☒ Transferring files and web pages over the internet
- D. ☐ Assigning IP addresses to devices

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What is the key difference between HTTP and HTTPS?

- A. ☐ HTTPS is used for file transfer, while HTTP is used for web browsing
- B. ☒ HTTPS encrypts data transmission, while HTTP does not
- C. ☐ HTTP is more reliable than HTTPS
- D. ☐ HTTP is faster than HTTPS

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Which protocol automates the assignment of IP addresses to devices on a network?

- A. ☒ DHCP
- B. ☐ TCP
- C. ☐ HTTP
- D. ☐ UDP

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Which of the following is NOT an advantage of TCP?

- A. ☐ Guarantees delivery of packets
- B. ☐ Packets arrive in order
- C. ☒ Lower overhead
- D. ☐ Ensures data integrity

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Which of the following applications typically uses UDP?

A. ☐ Email

B. ☐ File transfer

C. ☒ Online gaming

D. ☐ Secure online banking

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What does the padlock icon in a web browser's address bar indicate?

A. ☐ The website is using TCP

B. ☐ The website is using DHCP

C. ☒ The website is using HTTPS

D. ☐ The website is using HTTP

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Which protocol is responsible for translating domain names (e.g., [invalid URL removed]) to IP addresses?

A. ☐ DHCP

B. ☒ DNS

C. ☐ HTTP

D. ☐ TCP

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Which of the following is a potential security risk associated with using HTTP?

- A. ☐ Guaranteed data delivery
- B. ☐ Data encryption
- C. ☒ Man-in-the-middle attacks
- D. ☐ Automated IP address assignment

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